Stage 16N Report

Updated 2026-06-22

1 Overview

Stage 16N investigates restart-preserved cycle-jump behavior for the Chaboche UMAT Abaqus benchmark. The R4H diagnostic tested whether source-split restart files differ from restart records produced by long restarted replay runs. The R4I-R recovery rerun then repeated the cycle-280 source-split branch with durable lightweight copy-back and absolute reference CSVs.

2 Results

R4H completed six PBS/Abaqus jobs successfully with `Exit_status=0`. Long-replay restart records passed exactly: R4H1 at cycle 500, R4H3 at cycle 500, and R4H5 at cycle 750 all had zero global, local, and diagnostic S11 error. Short source-split restart cases remained inconsistent: R4H2 failed at cycle 500 with 11.83033% max primary local error, R4H4 was review-level at cycle 500 with 9.3860617%, and R4H6 was review-level at cycle 750 with 8.0369449%.

The R4I-R first wave finished by 2026-06-20 13:28 CEST and was absent from the live PBS queue on 2026-06-21 05:47 CEST. PBS history no longer returned records for jobs `1350601` and `1350602`, so the current classification uses scratch case status files, PBS stdout, Abaqus completion messages, and copied comparison CSVs. R4I-R1 passed exactly at cycle 500; R4I-R2 failed with the same source-split signature as R4H2.

Case	Mode	Endpoint	Status	Global %	Primary local %	S11 %
R4H1	Long replay 280–500	500	pass	0	0	0
R4H2	Source 250–281, restart 280–500	500	fail	1.1887403	11.83033	0.92936729
R4H3	Long replay 270–500	500	pass	0	0	0
R4H4	Source 250–271, restart 270–500	500	review	2.5314199	9.3860617	1.2392968
R4H5	Long replay 505–750	750	pass	0	0	0
R4H6	Source 500–506, restart 505–750	750	review	0.31896796	8.0369449	1.5702038

R4I-R first-wave recovery

Case	Source solve	Restart	Endpoint	Status	Global %	Primary local %	S11 %
R4I-R1	Deck-clone 250–281	280–500	500	pass	0	0	0
R4I-R2	Buffered 250–300	280–500	500	fail	1.1887403	11.83033	0.92936729

R4I-R1 shows that cloning/truncating the clean direct replay deck shape repairs the cycle-280 branch. R4I-R2 shows that a longer buffered generated source alone does not repair that branch; the dominant mismatch remains `HOLE_RING_SDV8_MAX`.

R4I-R second wave and deck-clone confirmations

R4I-R3 and R4I-R4 were submitted on 2026-06-21 through the guarded storage-gated scratch workflow. Both jobs requested one node with 16 cores, one MPI rank, 16 OpenMP threads, 90 GB memory, and 24 h walltime. R4I-R3 reproduced the generated-buffer review signature at restart 270 with 9.3860617% maximum primary local error. R4I-R4 reproduced the generated-buffer review signature at restart 505 with 8.0369449% maximum primary local error.

R4I-R5 then confirmed the deck-clone/truncate strategy for the 270–500 branch: deck-clone source 250–271, restart 270–500, zero global, primary-local, and S11 error at cycle 500. R4I-R6 and R4I-R6B did not classify the 505–750 branch scientifically. R4I-R6 failed during continuation `.stt` writing near cycle 731, and R4I-R6B failed earlier during source `.stt` writing near cycle 503. These are infrastructure/output-writing failures, not numerical method failures.

R4K controller preparation

A 2026-06-22 storage/source audit showed `/scratch9` at 99% use and found no visible clean cycle-505 restart source under `/scratch9/pr21vyci`, `/scratch/pr21vyci`, or the home project clone. The next prepared work is therefore split into two controllers: R4K1 runs the confirmed 250-branch exact deck-clone control `250 -> exact 280 -> 500`; R4K2 performs a 505-branch restart-source preflight and stops before Abaqus solve if no valid source is found. Both controllers include phase timing and lightweight evidence copy-back.

The controllers were submitted on 2026-06-22 through the guarded wrapper. R4K1, job `1350764.mmaster02`, finished with `Exit_status=0` after 06:01:15 walltime and 20:39:54 CPU time. Abaqus completed successfully, and the exact-control comparison at cycle 500 passed with zero maximum global error, zero maximum primary-local error, and zero diagnostic S11 error. This confirms the 250-branch deck-clone exact control `250 -> exact 280 -> 500`. The corrected R4K2 preflight is job `1350767.mmaster02`; it finished successfully in 6 s walltime and found one large non-R4I candidate from the earlier R4E exact-target scratch area, `stage16n_r4e2_exact_500_to_505_solve_506_to_750_exact_target_source`, with a 4.9 GB `.stt` and successful cycle-505 source `.sta`.

The follow-up storage-light validation controller R4K2B, job `1352353.mmaster02`, used that preserved R4E2 cycle-505 candidate directly. It did not generate a new source `.stt`; the scratch case linked the preserved source files, read `STEP=505, INC=54`, and continued cycles 506–750 with no continuation restart-write request. PBS recorded `Exit_status=0`, `Stageout_status=1`, 05:41:43 walltime, 19:16:27 CPU time, and about 3.38 average active cores out of 16. Abaqus completed through cycle 750, but the comparison was nonzero: 0.31896796% maximum global error, 8.0369449% maximum primary-local error, and 1.5702038% diagnostic S11 error. Therefore R4K2B is a scientific validation failure of the R4E2 505 candidate, not an infrastructure failure. The stage-out warning is recorded separately because lightweight evidence was recovered and committed.

Case	PBS job	Endpoint	Status	Global %	Primary local %	S11 %
R4K1	1350764	500	pass	0	0	0
R4K2B	1352353	750	review	0.31896796	8.0369449	1.5702038

3 Images

No R4H images were generated in this update.

4 Tables

The key R4H and R4I-R comparison tables are included in the Results section. Lightweight R4I-R evidence is stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4ir_restart_source_buffer_recovery/
```

Prepared R4K controller files are stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4k_deck_clone_controllers/
```

R4K1 lightweight result evidence is stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4k_deck_clone_controllers/
R4K1_250branch_controller/
```

R4K2B lightweight validation evidence is stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4k_deck_clone_controllers/
R4K2B_505candidate_validation_controller/
```

The earlier R4H CSV evidence is stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4h_restart_source_diagnostics/
```

5 Failures and Debugging

No R4H job failed at PBS/Abaqus level. The original R4I comparison failed because remote reference CSV paths were missing after extraction, and its scratch outputs were not recoverable. R4I-R fixed that workflow by staging absolute reference CSVs and copying lightweight evidence before comparison. The scientific failure now narrows to generated source-split consistency: R4I-R2 reproduces the R4H2 error even with a larger 250–300 source buffer, and R4I-R3/R4I-R4 show that generated post-target buffering does not repair the 270 or 505 branches. R4I-R6/R4I-R6B additionally exposed scratch `.stt` write failures on the 505 branch.

6 Conclusion and Next Steps

R4H, R4I-R, and R4K1 show that long-replay/deck-clone restart records are clean on the confirmed 250 branch, while generated short-source decks remain non-equivalent even with post-target buffering. Deck-clone/truncate is confirmed for 250–270–500 and for the exact-control path 250–280–500. The 505–750 branch now has a completed R4K2B validation of the preserved R4E2 cycle-505 candidate, and that candidate failed scientifically with the same nonzero 750-cycle signature observed in earlier source-split/review cases. Do not submit R4J9/R4J10 yet. The next useful solver path is the storage-light 250-branch true-jump controller after a fresh storage audit and resource-calibration decision; the 505 branch needs audit/redesign before further solver work.